Achieving global consensus on the strategic broadband access network — the Full Service Access Network initiative

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The Full Service Access Network (FSAN) initiative represents nearly half of the world's telephony lines. They have agreed upon a common broadband access system that can be applied to both the business and residential market sectors. The FSAN initiative has been responsible for harmonising the broadband requirements of many telcos; for example, it has produced the world's first asynchronous transfer mode passive optical network (ATM PON) specification which is being adopted world-wide. This paper describes the current activities within FSAN, the application of FSAN components and the future for FSAN.

1. Introduction

Imagine how much cheaper broadband networks could be if their components could be mass produced, say in the quantities needed for tens of millions of access lines, rather than today's typical few thousand line trials. This is the vision behind the Full Service Access Network (FSAN) initiative. A three-year-old project, that involves fourteen of the leading telecommunications network operators, supported by ten of the major telecommunications equipment manufacturers. The aim is to create a common requirements specification for an access system supporting a full range of narrowband and broadband services for a market that covers one third of one billion lines of telephony.

The aim of this paper is to communicate the achievements of the FSAN initiative, describe the common access system and explain the potential application areas of this technology. The telcos currently involved in the initiative are Bell Canada, Bell South, BT, Deutsche Telekom, Royal PTT Nederland, France Telecom, GTE, Korea Telecom, NTT, SBC, Swisscom, Telecom Italia, Telefonica, and Telstra.

Within FSAN it was recognised that the needs of individual telecommunications operators differ due to different regulatory, business, and structural environment in each country. However, sufficient similarities exist in requirements for future access networks to enable significant benefits to be achieved through adopting a common requirements specification. The common system is based around a broadband fibre feeder system, ATM PON (passive optical network), and an xDSL (digital subscriber line) system, shown in Fig 1. The exact DSL system used depends upon where the optical system is terminated, e.g. in the local exchange, cabinet, kerb or home. Hence this broadband access system can support a range of access architectures; this flexibility is fundamental to the consensus achieved in FSAN. Dickie and Mackenzie [1] describe the use of asymmetric digital subscriber line (ADSL), one of the topology options shown in Fig 1.

2. The need for broadband

It is anticipated that FSAN will provide a broadband solution to both the residential and small/medium enterprise markets. Table 1 captures the incumbent operator's position in the emerging broadband environment. The existing revenue streams are under increasing pressure, and the nature of network usage is also changing, e.g. the rapid growth of the Internet. FSAN provides a broadband access solution that builds upon the operator's strengths through using the existing infrastructure; it also tackles some of the weaknesses by minimising capital investment. FSAN is flexible in its topology through a range of reuse options, thus managing the threats. FSAN is enabling operators to jointly take advantage of the broadband opportunity through minimising the risks associated with the investment.

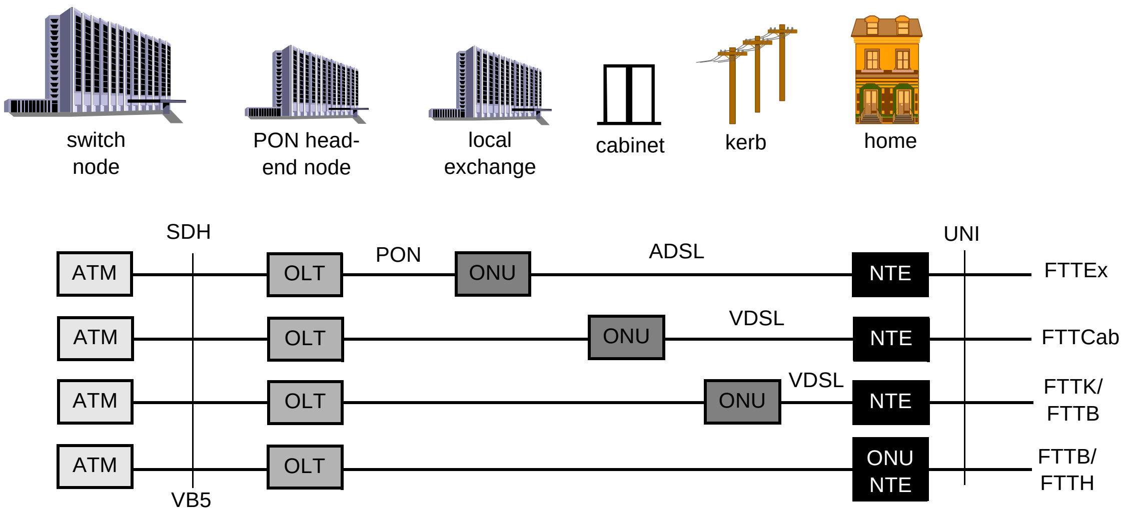

Fig 1 Common network elements.

Table 1 The telecommunications operator's position in the emerging broadband environment.

Strengths Existing ubiquitous infrastructure. Organisation, e.g. sales, marketing, maintenance, able to support new broadband services. Reputation. Excellent brand awareness.	**Weaknesses** Existing infrastructure with ongoing depreciation charges necessitates reuse. Capital constraints limit investment. POTS is devaluing so need new revenue streams. Market will be strictly regulated.
Opportunities New revenue streams — teleworking, Internet and multimedia. Lower cost of providing POTS. Technology enabling broadband multimedia delivery to the home.	**Threats** Uncertainty in demand and regulation. Competitors already providing broadband. Low service take-up. High burn.

3. FSAN structure

The FSAN initiative has been through four phases, each lasting roughly one year, the results being presented to the industry at public conferences; they are also available on the World Wide Web [2]. At first the initiative focused on identifying how cost reduction could be achieved. Several important system components were identified, and with a common specification it was believed that cost reduction would be possible. The global access product achieves cost reduction through two effects — competition in supply which reduces margins, and each supplier producing a greater volume of the common system so that the learning curve effect has a greater impact on production costs.

As the initiative has progressed the focus has moved away from bit level specifications to management/control plane issues and service-specific aspects, e.g. how fast channel changing is done for switched digital broadcast TV (see section 5.1).

Figure 2 shows the relationship between the different groups within FSAN. The workgroups add content to the specification. The chapters use the specification to achieve their aim of available systems meeting the FSAN specification. The achievements of the groups shown in Fig 2 are described in the following sections. The objective is to complete the outstanding technical issues by the end of 1998, and provide a milestone upon which the manufacturers can focus their development programmes.

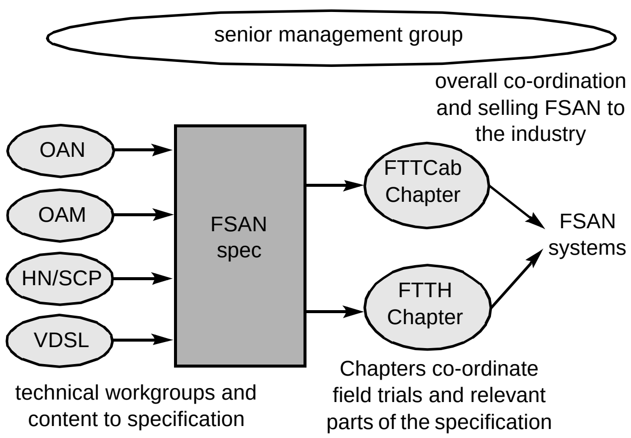

Fig 2 Structure of FSAN.

4. FSAN Chapters

4.1 FTTCab Chapter

This Chapter focuses on commercial aspects such as equipment availability, ensuring all FTTCab aspects are covered in workgroups, joint evaluation of RFPs, and joint testing and trials. The members of this Chapter are Bell Canada, British Telecommunications plc, Deutsche Telekom, France Telecom, GTE (US RBOC), Korea Telecom, SwissCom, Telecom Italia and Telstra.

4.2 FTTH Chapter

This Chapter is focusing on technical aspects specific to FTTH and FTTB (small and medium business customers), e.g. ONU environment, powering, size, etc. It provides input to technical workgroups, and is not yet focused on joint trials. Its members are Bell South, France Telecom, NTT, British Telecommunications plc, Deutsche Telekom, Royal PTT Nederland, SBC, Swisscom and Telecom Italia.

5. FSAN Workgroups

5.1 Service Capabilities and Performance WG

The mandate of the SCP group is:

- to define and specify the capabilities required of FSAN access networks to support the required services,
- to determine dimensioning/sensitivity and performance requirements of an FSAN,
- because FSAN is access-centric, the focus is on bearer services, i.e. transport-related.

The current work areas of the group are:

- switched digital broadcast (SDB), which covers the following issues:

  — fast channel changing (zapping) protocol,

  — replication in the access to conserve capacity,

  — customers having access to multiple broadcast content providers simultaneously;

- real-time control of access network resources, which covers the following issues:

  — full signalling (Q.2931) versus possible new streamlined low-level protocols,

  — resource management of network termination is a key issue;

- integrated POTS/voice, multiline voice, for example:

  — lifeline, toll quality,

  — non-lifeline, lesser quality,

  — other classifications based on availability;

- handling of IP capabilities and Ethernet/10BaseT user interfaces, with a focus on:

  — multiservice, multiterminal support,

  — concentrating on determining operator requirements for input into standards bodies.

Taking the switched digital broadcast issue as an example, the group has decided not to provide a special solution for zapping between TV channels provided over the access system, but follow the VB5 architecture, as is used for all other services over the access system. Figure 3 shows the architecture for zapping. A channel change request, zapping, comes from the STB (set-top box). It is not terminated in the access, for many operational and commercial reasons, e.g. this avoids the need for the access network to contain customers' personal information. It enables the zapping protocol to be independent of the access network — only the service provider and STB need to be aware of the protocol. Through the bearer channel control (BCC) protocol the access receives the instruction for the channel change.

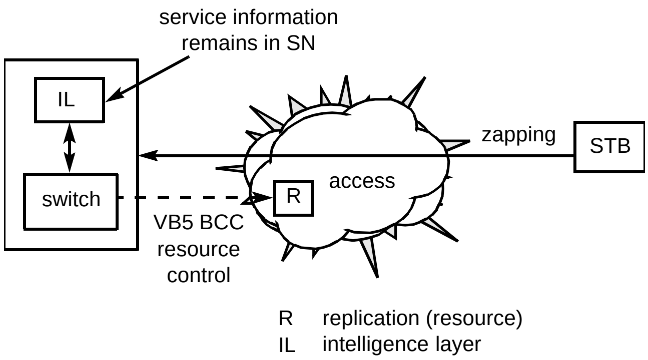

Fig 3 Architecture for zapping.

5.2 Operations, Administration and Maintenance WG

The Operations, Administration and Maintenance (OAM) WG was set up to look at the operational aspects of the FSAN and to agree on a common set of management functions. A framework for developing a common set of requirements was quickly established which included the following steps:

- understand the operational and service requirements (what are we trying to deliver),
- derive the management requirements and develop a reference architecture (how it is to be managed),
- develop specifications to ensure standardisation in the supplier industry.

A key decision of the group was to look at the problem from an end-to-end perspective which led to the group adopting a set of high-level processes from which it was possible to agree on a set of common operations functions. A number of high-level processes are described in the FSAN OAM requirements document [3], including service provision, network repair, and planning and engineering. Each of these contain elements of customer handling, work management, installation, billing, etc.

Another key agreement of the group was to reuse as many existing standards as possible in order to make rapid progress. When it was clear that an architecture was required to progress discussions on the requirements, the basic architecture described by the Telecommunications Management Network (TMN) was adopted and extended to develop an FSAN target architecture as shown in Fig 4.

Over the first three phases of the FSAN initiative the group has defined requirements on the following aspects of operations and management:

- high-level processes,
- operational requirements of the equipment — modular design, simple visual indications, self-configuration, accurate fault diagnosis,
- management requirements — management architecture, interfaces and functions,
- platform requirements — scalability, throughput, performance, operating system, security, availability,
- data network communications — types of data network to be supported,
- test equipment — automated testing, reduction of reliance on network testing,
- VDSL — operational requirements of VDSL link (management of physical layer),
- ATM layer requirements — OAM flows,
- information model — based on ATM Forum M4 and ITU-T models.

Most of this work has been completed in nine meetings since work began in late 1995 with a high level of commitment and support from both operators and suppliers.

In the next phase the OAM group will be progressing the following areas of work:

- define information model details,
- collaborate with the Optical Access Network (OAN) WG on management requirements for the management channel on OLT-to-ONU/ONT interface,
- support programme to standardise FSAN requirements,
- explore common management requirements for higher layer functions.

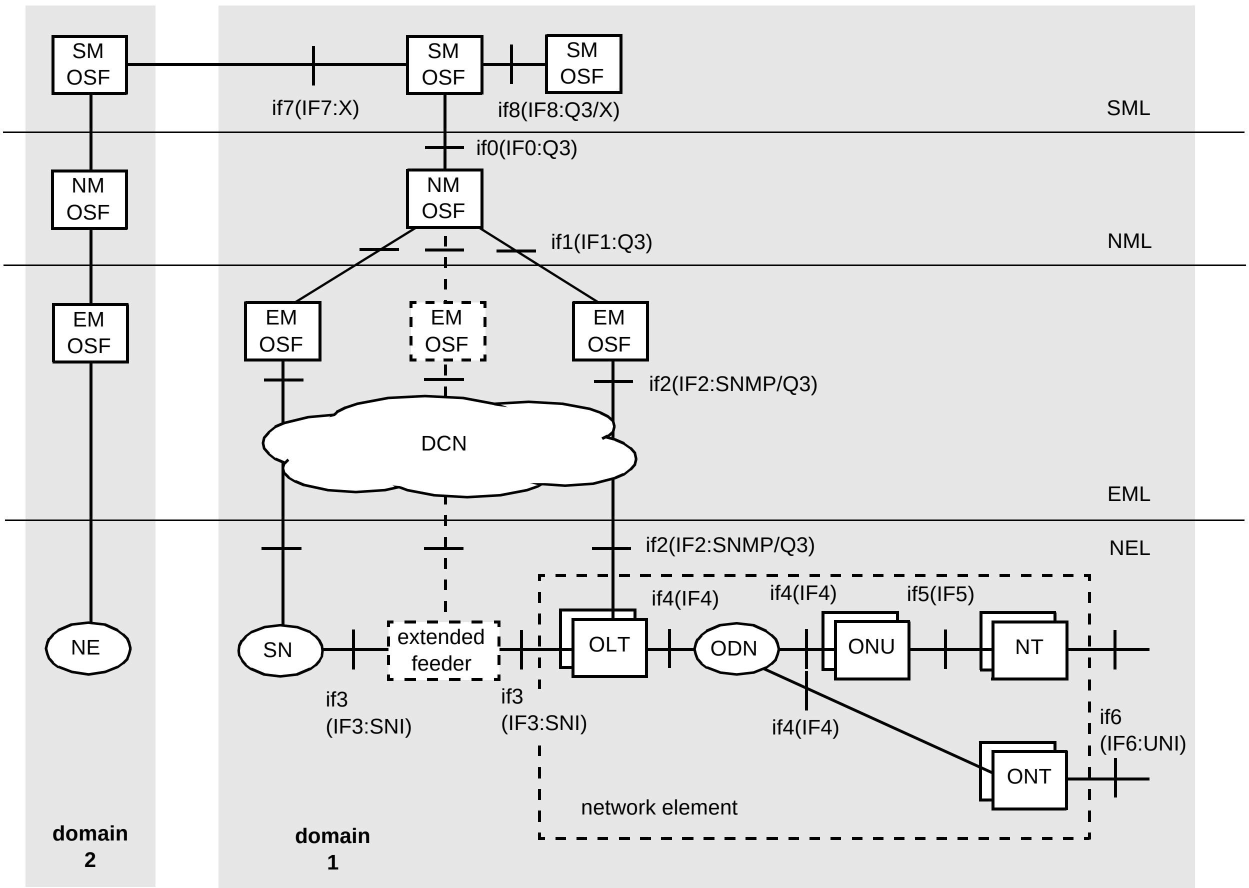

Fig 4 FSAN target management architecture.

The current sets of requirements cover the NE and EM layers of the TMN. Further work is needed to extend the FSAN management requirements to the network management (NM) and service management (SM) layers.

The FSAN requirements have been discussed at a meeting of the appropriate ITU-T work group in March 1998 and a draft ITU-T document containing the FSAN requirements has been produced. The OAM group has completed work on the definition of an information model for interface IF1 (see Fig 4) which became available in July 1998. In addition, both the requirements and the information model were input into ETSI in September 1998. The aim is to have agreed recommendations by the end of 1999 following the normal procedures of the ITU-T and ETSI. The group is also looking at adoption of the FSAN requirements as an ANSI standard but discussions are at an early stage.

It has only been possible to give a high-level view of the work of the OAM group in this section. The reader is referred to the OAM requirements document [3], and other FSAN papers [4—6] for more details of the areas described above.

5.3 Optical Access Network

This group has created the world’s first ATM PON specification. It has been presented to and accepted by the ATMF [7], ITU [8] and ETSI [9]. The specification focuses on physical layer and transmission convergence layer (shown in Table 2), following the reference configuration of ITU-T Recommendation G.PONA (G.982) [10]. The system has been designed to support all the configurations shown in Fig 1.

Table 2 PON layers and functions.

<table>
<tr><td colspan="3">Circuit layer</td><td>Translation and maintenance</td></tr>
<tr><td colspan="3">Path layer</td><td>I.732</td></tr>
<tr><td rowspan="3">Transmission Media Layer</td><td rowspan="2">Transmission convergence layer</td><td>Adaptation</td><td>I.732</td></tr>
<tr><td>PON transmission</td><td>Ranging
Cell slot allocation
Capacity allocation
Privacy and security
Frame alignment
Burst synchronisation
Bit/byte synchronisation</td></tr>
<tr><td colspan="2">Phyical media layer</td><td>E/O adaptation
WDM
Fibre connection</td></tr>
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There are two PON options, a symmetric 155 Mbit/s, and an asymmetric 622 Mbit/s down to the customer and 155 Mbit/s from the customer. The PON also has a minislot capability to support voice, STM (synchronous transfer mode) services or dynamic capacity over the PON. The PON’s dimensions are a 20 km reach, and a maximum optical split of 32. The system can operate on either one fibre using WDM, or two fibres.

Figure 5 shows the frame structure of the APON. It is ATM cell based. A downstream frame is made of 56 cells, which includes two PLOAM (physical layer operations, administration and maintenance) cells. The PLOAM cells contain the grants for the upstream slots, and messaging for physical layer functions, e.g. to instruct an ONU to start

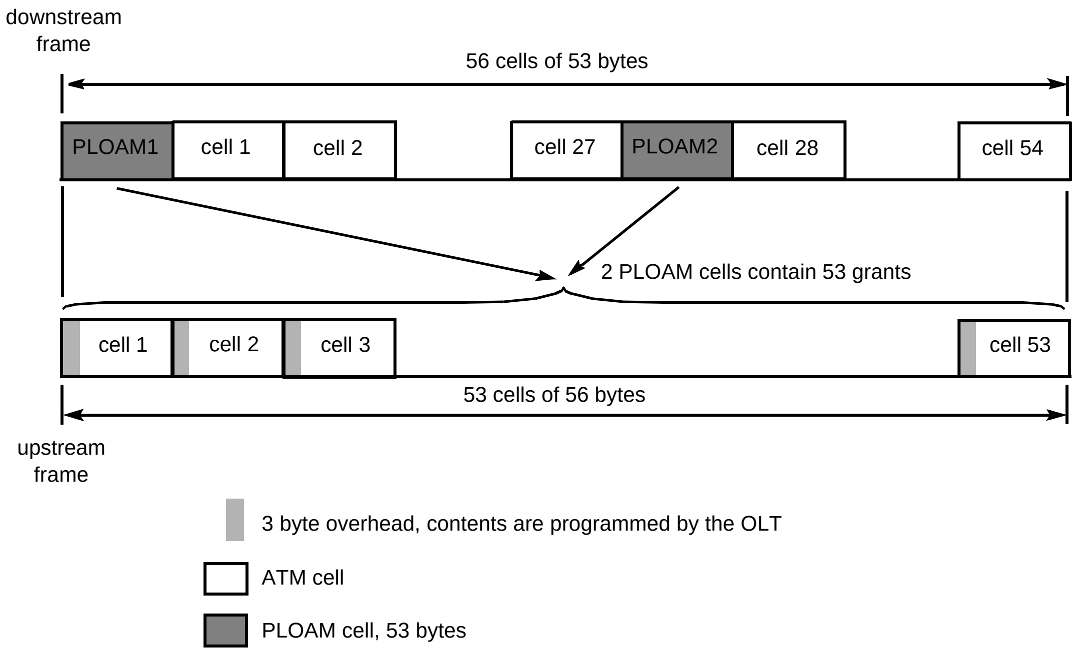

Fig 5 The APON frame structure.

ranging. The upstream cell has a 3-byte header, to cope with the multiple access nature of the upstream transmission path. Because of the overhead, the upstream frame is made up of 53 upstream slots. The mechanism for allocating the slots, the media access control protocol, is not defined. This was left open to enable vendors to innovate. The OLT is the controlling device. The ONU responds to the OLT's commands, and therefore the PON interface can still be manufacturer independent.

5.4 Very high rate digital subscriber line

VDSL technology can deliver data at multi-Mbits/s over the unscreened, twisted telephone wires originally intended for bandwidths of between 300 Hz and 3.4 kHz. This is due to remarkable advances in digital signal processing technology which allow the implementation of sophisticated digital modulation and equalisation schemes. In fact, the challenges are now mainly of an analogue nature — channel attenuation, noise and spectrum management. More details on the use of DSL in the access network can be found in Foster et al [11].

It is important to note that VDSL is required to operate on underground as well as overhead distribution cabling. This imposes some difficult requirements in order to control unwanted RF emissions, particularly for overhead distribution cabling. VDSL will also be required to operate in the presence of bridged taps — principally a problem for North American telcos.

General description of VDSL

Figure 6 shows a simple architectural model of VDSL. Since the broadband and narrowband services should be able to share the same metallic distribution cable, the broadband services are introduced at frequencies well above POTS or ISDN-BRA. Figure 7 shows the VDSL signal placed well above the band occupied by POTS or ISDN-BRA.

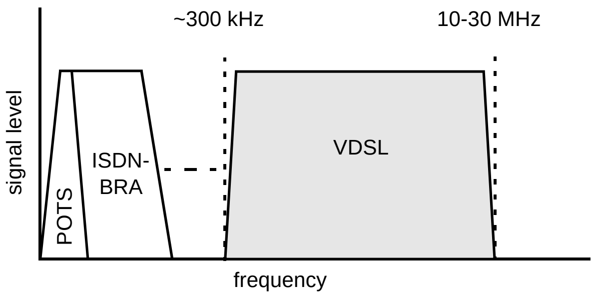

Fig 7 Frequency separation of VDSL.

The VDSL standardisation process

The introduction of VDSL transmission systems is dependent on harmonisation of network operators' requirements to produce a large common market. The remit of the FSAN VDSL working group has therefore been to identify the key requirements which are common to all telcos, and downstream these requirements to the standards forums — notably the European Telecommunications Standards Institute (ETSI), the American National Standards Institute (ANSI), and more recently, the VDSL Study Group within the ADSL Forum.

The FSAN VDSL WG

The FSAN VDSL Working Group has made a significant contribution to the VDSL standards process by achieving telco consensus on the key systems requirements for VDSL. This work was published in some detail at a workshop in Atlanta in March 1997 [12]. The work has now been successfully downstreamed such that the current version of the ETSI draft technical specification for VDSL [13] should

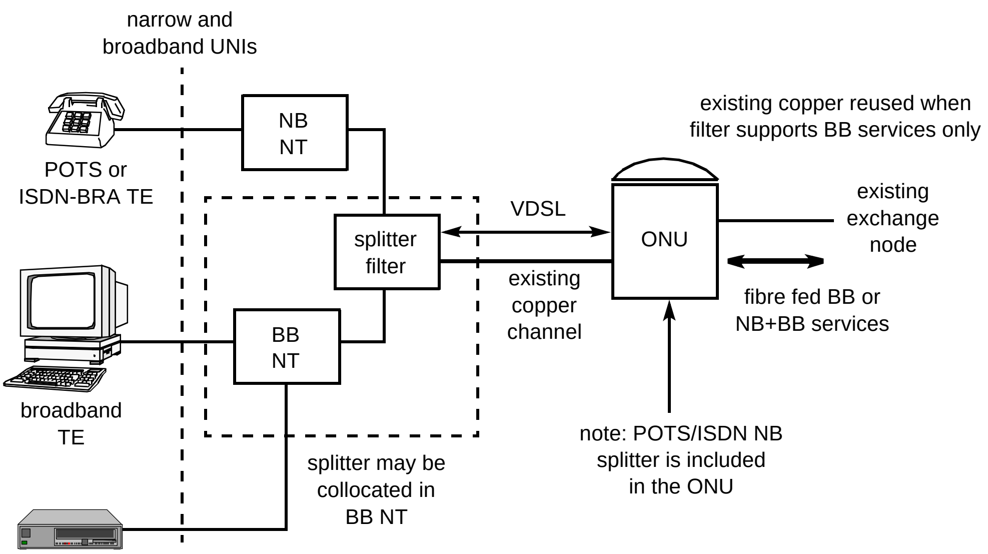

Fig 6 VDSL concept.

be viewed as the primary reference source for FSAN VDSL requirements at this time.

ANSI has adopted much of the ETSI work, with the FSAN requirements being actively progressed by the North American members of FSAN, notably GTE and Bell Canada.

During 1997-8, the group has been addressing the complex technical issues of noise model and spectral compatibility for VDSL. Spectral compatibility is a fundamental issue for telcos — to ensure that different xDSL systems can co-exist in the same cable infrastructure, maximising their performance, while minimising their impact on neighbouring systems due to crosstalk. Because spectral compatibility is a systems issue which is network dependent and which transcends the boundaries between xDSL systems, the group has remained telco-only to date.

VDSL noise model

A major achievement of the group has been the definition of a noise model for VDSL that was derived during a telco-only meeting in Paris in November 1997. The noise model is required to enable telcos to benchmark vendor implementations and to verify that systems will work in real networks.

At the Paris meeting, different telco xDSL deployment scenarios and cable topologies were modelled in real time to identify a small set of noise profiles that could be applied to all telcos. A powerful enabler for consensus at the meeting was the availability of two independent laptop computer models which could be used to check results in real time. It was found from modelling that all basic telco scenarios could be fitted to only four noise profiles — a result that was not obvious beforehand. These profiles have been accepted by ETSI [14] and have been provisionally adopted for inclusion in the ANSI standard.

This is believed to be the first time that a consensus noise model has been defined so early in the standards life cycle for an xDSL system, demonstrating the value of this method of reaching consensus.

A key insight from the noise model work was confirmation, previously highlighted by Nortel [14], that the spectral mask for ADSL required modification to minimise impact on VDSL capacity. This has been progressed in ANSI by GTE on behalf of the FSAN telcos. It was also noticed that existing standards relating to the spectral masks for ISDN-BRA were not adequate to prevent new systems polluting the VDSL spectrum above 1 MHz. Fortunately, measurements on installed ISDN-BRA systems (both 2B1Q and 4B3T) have confirmed that existing systems are VDSL friendly; but the ISDN standard is currently under review in ETSI, and vigilance needs to be maintained to prevent any relaxation of the mask, which would open the door to spectral pollution from future variants of ISDN-BRA systems.

VDSL spectral compatibility

More work is now required by the FSAN telcos to define the spectral bounds and duplexing scheme for VDSL before vendors have sufficient information to develop spectrally compatible systems.

Although spectral masks for VDSL have been agreed in ETSI, these only detail the major features, such as notching and maximum power. To ensure that systems are developed which are spectrally compatible with installed xDSL systems such as ADSL, telcos must provide more detail to vendors on the spectral bounds for upstream and downstream VDSL transmission, and evaluate the system implications of the different duplexing techniques.

This is an extremely complex problem to resolve because VDSL capacity is a complex function of topology, duplexing scheme and crosstalk. The final choice will depend on which engineering compromises telcos are willing to adopt, and this depends on the ability of the telcos to model their scenarios for VDSL upstream and downstream capacity.

To resolve this problem, a common formula and methodology for calculating VDSL capacity in different telco scenarios was agreed at a meeting in Bern in February 1998. The models have been verified using a single common scenario and will be used by individual telcos to evaluate their own xDSL scenarios in order to develop a group insight into this complex problem.

Next steps

If, as is hoped the group achieves consensus on the spectral bounds and duplexing scheme for VDSL, it will have achieved its primary aim of defining the spectral parameters and systems requirements for VDSL. Vendors will then have enough information to confidently invest in the development of FSAN-compliant VDSL equipment. If consensus is not achieved, a degree of additional flexibility will be required in vendor equipment to guarantee that a particular VDSL implementation will operate reliably in all telco networks world-wide — the overall aim of the FSAN initiative.

Telcos ultimately require that systems are not just spectrally compatible, but that a modem from one vendor will interwork with that from another vendor. This is primarily the task of international standards.

Alleviating the standards burden

The world spotlight is on xDSL as the technology that will provide universal broadband access as we enter the next millennium. This has significantly increased the pressure on xDSL technical experts to attend meetings all over the world. The FSAN initiative provides an opportunity to alleviate some of this pressure (and save costs) by sharing technical perspectives, and co-ordinating attendance at meetings.

Following the FSAN summit in Venice in March 1998, the remit of the VDSL Workgroup has been extended to encompass general liaison on all xDSL issues of common interest to the telcos, in particular to maintain vigilance, and co-operate to ensure that xDSL systems are spectrally compatible with each other, and, through technical dialogue, to resolve the difficult engineering issues presented by, for example, the Splitterless ADSL concept (DSL Lite).

5.5 Home network

The work of FSAN is very much service driven. Since services are by definition end to end, then FSAN must ensure that it considers a complete end-to-end system description, which includes the customer environment. This is something of a change from the traditional telco view, where responsibility (and possibly interest) ended at the NT. Although there may still be a limit to which operators can dictate what occurs in the customer environment, they can at least ensure that suitable customer systems are specified. Three aspects of the customer environment need to be considered:

- the home network,
- the customer interface,
- CPE functionality.

Decoupling of the home network

The basic problem that the home network needs to address is how to distribute the access network services throughout the home, and provide multiple simultaneous attachments to different services, while retaining the quality of each service. The solution to this problem involves issues that range from architecture to infrastructure. The basic FSAN HN architecture is shown in Fig 8. Two key features are shown:

- the access network is decoupled from the HN by means of an active NT,
- the HN is essentially a LAN rather than simply being an extension of the access network — however, it must be a LAN that can support the required service mix.

This decoupling means that different and appropriate transmission systems can be used in the access and home environments, and that the same HN can be used with a variety of access network types. The second main feature is

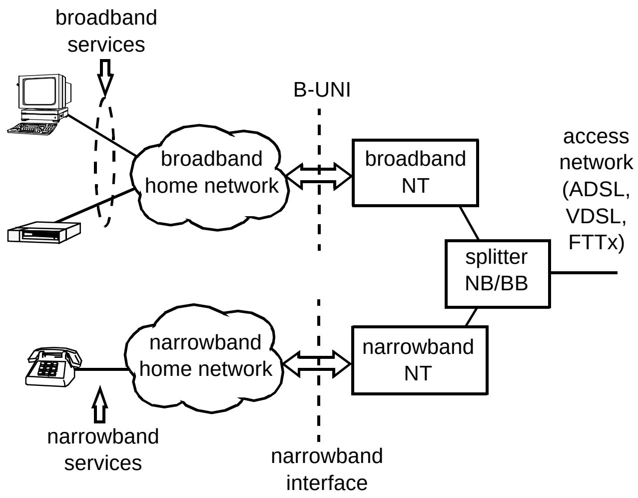

Fig 8 Home network architecture.

the separation of the broadband home network from the existing narrowband one.

After consideration of all the existing and developing technologies, the following solution was agreed:

- a point-to-point star architecture,
- newly installed category 5 twisted pair cable,
- 50 m reach between nodes,
- basic network functionality incorporated into the NT,
- enhanced functionality provided by means of a separate hub.

The basic functional elements of the home network are shown in Fig 9. The NT1 terminates the access line system (xDSL or fibre) and may contain other functionality, in particular basic multiplexing and demultiplexing. Multiple (customer-side) interfaces may be presented by the NT1. The NT1 functionality will be physically located in the B-NT. Full switching requires the addition of an NT2. This may be located in a separate physical device, or could be incorporated into the B-NT; however the degree to which this incorporation is allowed may be the subject of local regulation. A separate NT2 would generally be regarded as CPE. The CPE end devices can connect directly to the NT1, NT2 or both, depending on the implementation.

The customer interface

The general framework of FSAN is based on the fact that the transfer mode of the access network needs to be based on ATM to provide the controlled, full service mix. In order to retain this full service capability to the user, ATM needs to be continued right through the home network to the CPE. This also avoids a complex interworking function in the NT. The problem is that the home environment is rather different from the business environment with regard to the required transmission and EMC performance. However, the ATM Forum have developed a PHY standard specifically

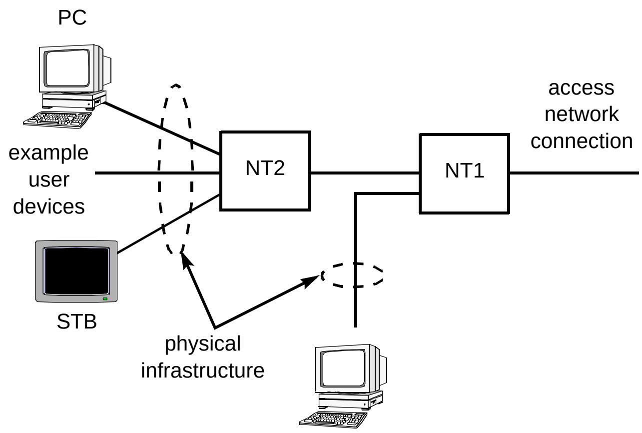

Fig 9 Basic functional elements.

designed for the residential environment. This is based on (and backward compatible with) the original ATM25, but provides:

- a higher line rate mode (51.2 Mbit/s),
- improved EMC properties,
- physical layer OAM.

This is the chosen FSAN customer interface. It provides sufficient bandwidth to support future HDTV services as well as allowing for more extensive customer networks. The 51 Mbit/s mode of this PHY must be supported in FSAN systems, but it will automatically interoperate with devices that conform to the original ATM25 specification, thereby providing backward compatibility with some early, existing deployments.

However, for some specific services, there may be a case for terminating ATM before the CPE (e.g. in the B-NT or a separate terminal adapter) and providing interworking to another network type. The only example of this alternative approach considered by FSAN is Ethernet interworking. A key issue with Ethernet presentation is where the (layer 2) Ethernet protocol is terminated, i.e. where the initial routeing functionality is carried out. The preferred FSAN approach is to do this in the home for reasons of scalability and security; this functionality could either be incorporated in the NT or as a separate, conventional small router. 10BaseT was not designed for the residential environment and does not have desirable EMC properties. The FSAN-preferred Ethernet interface type is therefore 100BaseTX. However, in order to provide compatibility with 10BaseT PC interface cards, where 100BaseTX is implemented in the B-NT, it must support dual mode 10/100 Ethernet operation.

CPE functionality

A range of appropriate broadband CPE must be available, together with attractive service offerings, to ensure that a new access network is fully utilised; lack of CPE and services inhibited the early take-up of ISDN. An objective of the FSAN members is to gain consensus on the core requirements for customer premises equipment and thus ensure compatibility and interoperability with FSAN networks and services. This would also enable the member companies to leverage commercial advantage in the supply of CPE either through the normal retail chain or via direct procurement.

The term CPE embraces PCs with network interface cards (NICs), network computers (NCs) and set- top boxes (STBs).

The consumer electronics industry will be invited to comment and input, as necessary, to these core requirements since FSAN’s goal is to influence their ‘roadmap’ for future products. Key CPE features already identified which need to be addressed include:

- user-friendly CPE which is truly ‘plug-and-play’,
- user-to-network flows terminating in the access network,
- user-to-user flows terminating beyond the access network,
- traffic shaping,
- APIs/middleware,
- standardised software download,
- configuration management,
- security.

For residential services, the intention is that the core CPE requirements developed by FSAN may be added to next-generation consumer electronic equipment such as digital satellite receivers, digital terrestrial TV receivers, DVD players and games consoles. This would then enable the consumer to access multiple services from a single STB.

5.6 Infrastructure

The implementation of VDSL fibre-to-the-cabinet has traditionally taken the form shown in Fig 10. The VDSL electronics are housed in an above-ground cabinet, and positioned close to an existing copper primary cross-connection point (PCP) in the network. The VDSL access point (VAP) is linked to the exchange via fibre. Copper ‘tie’ cables are used to link the VAP to the PCP in order that broadband service can be overlaid on the existing D-side copper network.

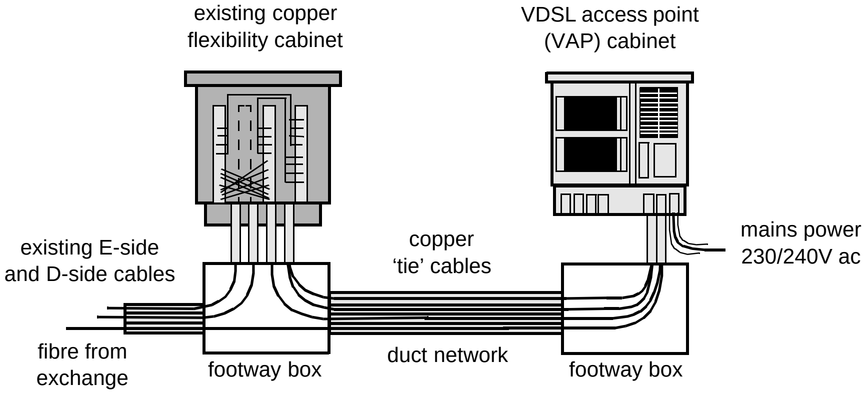

Fig 10 Traditional above-ground VAP FITCab scenario.

Placing active electronics in a hostile external environment presents major design challenges. The combined requirements for line capacity and the associated power dissipation, plus the auxiliary facilities for copper, fibre and power can result in a physically large VAP cabinet with a complex and expensive cooling system. These issues have been the focus of in-depth discussion for the FSAN infrastructure group, where telcos and suppliers are endeavouring to reach a common consensus on the design requirements for a FTTCab solution. These discussions are progressing well, but have clearly highlighted that the various telcos have significantly different requirements and constraints on the design and use of active street furniture. Generic requirements for an above-ground solution are well advanced and are expected to be issued shortly. However, restrictions on certain telcos to not use physically large, or in some cases any, above-ground cabinets have resulted in FSAN also proposing an underground solution.

Figure 11 shows the underground concept that employs 8 or 16 line, sealed-for-life modular VAP enclosures which can be located in existing underground footway boxes. Each VAP module should provide suitable connections for primary power, fibre, copper 'tie-cables' and the option for battery back-up. The advantages of an underground system have been identified as:

- less hostile environment for temperature extremes,

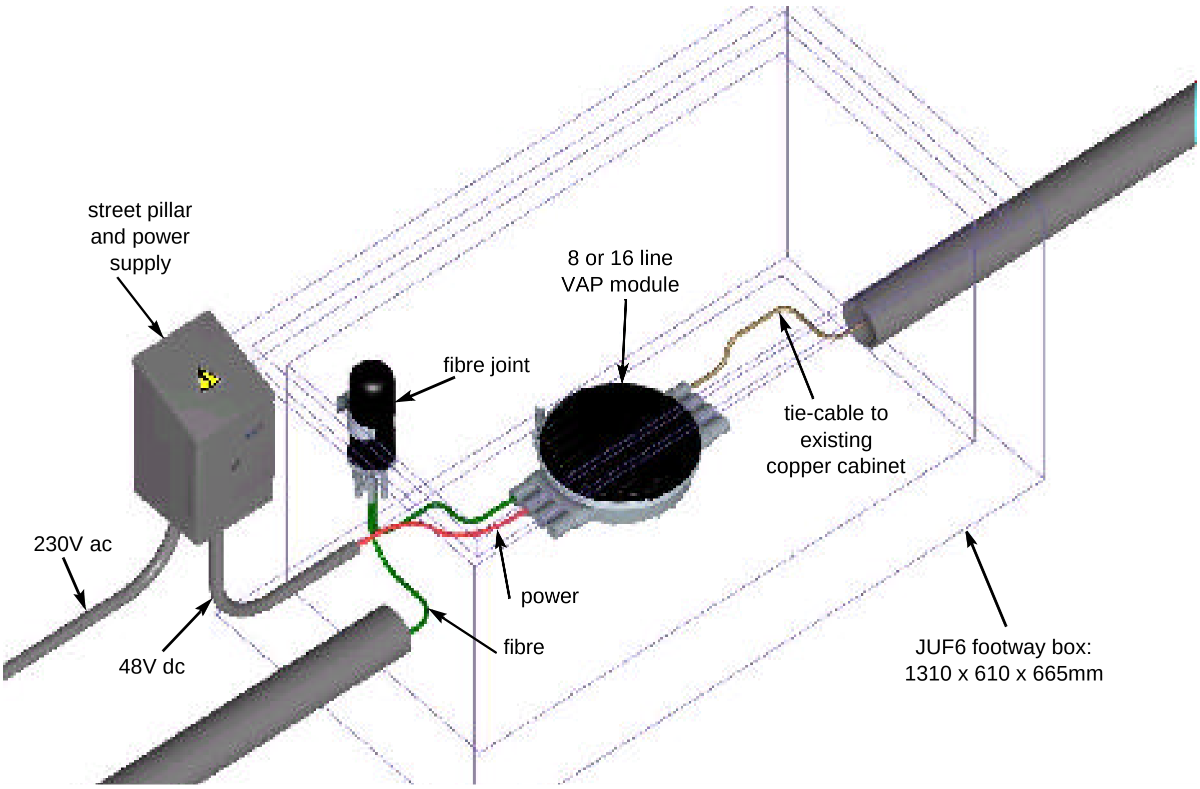

Fig 11 Underground modular VAP proposal.

- improved electronics and battery reliability,
- less prone to EMC radiation damage and emissions,
- more aesthetically pleasing,
- less vulnerable to accidental or deliberate damage,
- savings in enclosure costs compared to cabinets.

The underground system is designed to be highly modular allowing extra modules to be added to satisfy growing demand, as demonstrated in Fig 12.

This is also intended to be highly flexible to satisfy variations in telco requirements, and has the potential to be significantly less expensive than a cabinet solution. The proposal is attracting major interest from the telcos and suppliers within FSAN who are currently discussing the development issues and potential applications.

6. Next steps within FSAN

The latest results in FSAN were presented at ISSLS'98. FSAN is a temporary body. After the initial trials are complete, and the FSAN specification [2] is updated to reflect both the learning of the trials, and the systems for commercial deployment, FSAN's job is complete. The systems are available to all, because the specification is public; hence volumes are up and prices are down. This is the FSAN vision.

7. Conclusions

The keys to FSAN's success are a political will to facilitate the creation of a common access system specification to enable vast economies of scale, and a highly flexible technology, PON and xDSL, that can meet the disparate requirements of the telecommunications operators involved in the initiative.

The FSAN chapters are aiming at accelerating the availability of FSAN systems, through a co-ordinated approach to testing and trials.

The SCP group was only recently created; its mandate is to identify capabilities in the FSAN access network to support a full set of services.

The FSAN VDSL working group has established a powerful discussion and modelling methodology for resolving the complex problem of xDSL spectral compatibility, and rapidly achieving consensus through mutual insight. This forum is unique in the world in that it contains xDSL technical experts from the core development teams of the major telcos, working harmoniously together to resolve telco-specific transmission problems, thereby accelerating standards through advance consensus.

The OAN group has created the world's first ATM PON specification. It has been presented to and accepted by the ATMF, ITU and ETSI. The system has been designed to support the requirements of all telcos.

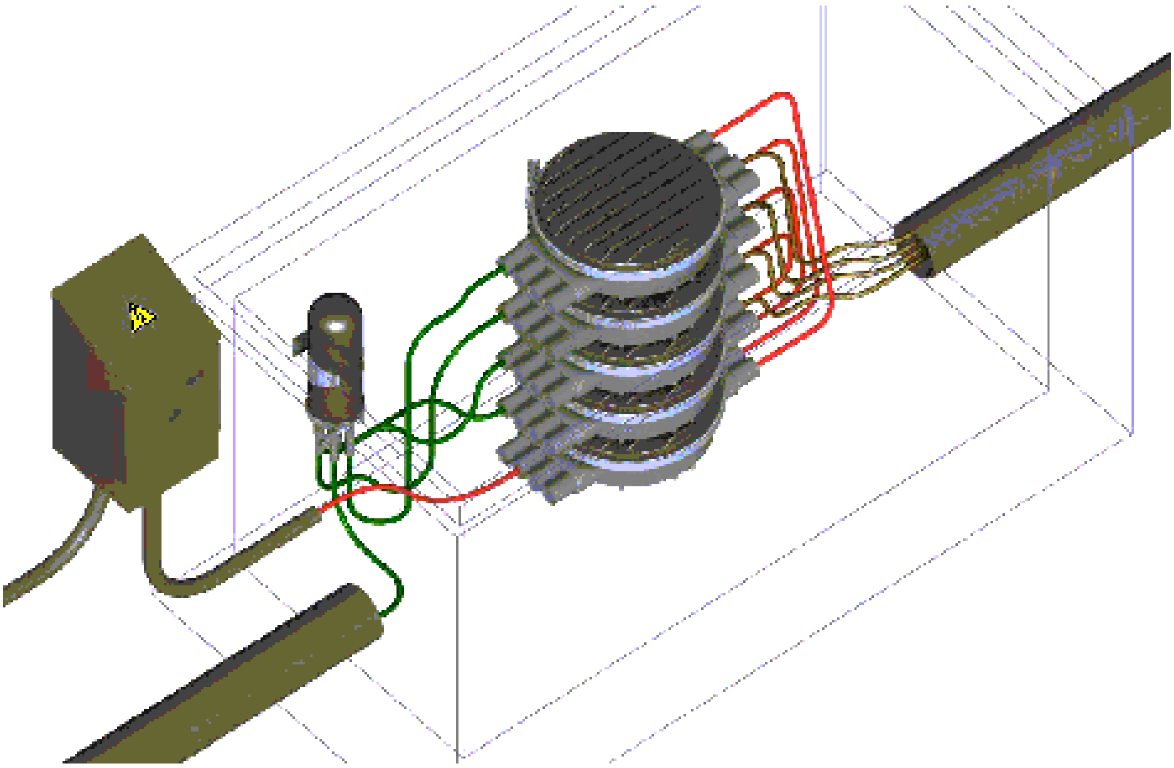

Fig 12 The VAP system allowing for growth in demand.

FSAN has specified the world's first truly global broadband access system through the dedication and expertise of the people involved in the initiative. The challenge now is how to realise the benefits of the consensus achieved within FSAN.

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Alan Quayle has two Masters degrees in electronic systems engineering and business administration. Since joining BT he has worked on a wide range of projects, from optical system design to developing VOD systems.

Currently he is leading work identifying broadband opportunities and the access systems to exploit them. He represents BT in the ATM Forum and Full Service Access Network initiative. He is a member of the IEE and IEEE.

Bob Bissell joined BT in 1973. He was involved with the development of electret microphones, BT's first 'in-house' designed electronic telephones and the introduction of new public payphones. He then led the interactive multimedia services terminal applications team, supporting the interactive set-top box development for both the interactive TV technology and market trials. He is currently a technical group leader within the Products and Services Engineering Centre at BT Laboratories leading a team working on customer premises equipment including interactive set-top boxes for off-air broadcast and broadband network services.

Don Clarke joined BT Laboratories in 1973. Based in Ipswich, he worked on development of the Pathfinder digital exchange — a forerunner of System-X. He graduated in computer systems from the University of Essex in 1981. Until 1995 he led the fibre systems group responsible for the development of narrowband passive optical networks, and holds a number of patents relating to that technology. In 1995 he transferred to the copper access group and is project manager for the ADSL and VDSL development activities. He has chaired the international FSAN VDSL working group for the past two years. He is a member of the IEE and IEEE.

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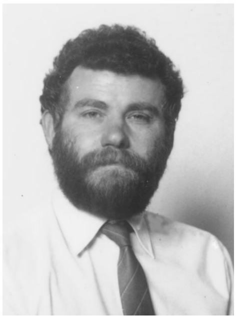
Jeff Stern is development manager, fibre and broadband networks at BT Laboratories, Martlesham Heath, Suffolk. He graduated from Imperial College of Science and Technology, University of London, with BSc(Hons) in Physics. In the earlier part of his career he worked on a broad range of topics in optical communications and played a key role in BTs pioneering work on single mode optical fibre systems in the early 1980s. During the late 1980s he led research work at BTL aimed at introducing optical fibre into the local network, including the development of the passive optical network (PON) concept. Current responsibilities include the development of long-haul submarine, long-haul terrestrial and local access fibre systems and the co-ordination of development activities aimed at achieving a future broadband network infrastructure. He has played a leading role in the FSAN initiative from its outset and is chairman of the FTTCab Chapter.



David Thorne holds a BSc in Physics from Imperial College, London, and a PhD in Radio Astronomy from the University of Manchester. He began his career with Marconi Elliott Avionics, and then moved briefly to ICL before embarking on 10 years in the instrument science division of the Royal Greenwich Observatory. He joined BT in 1989, initially working on the development of novel fault-finding techniques for the access network. He then moved into the private data network area and currently leads a team which is concerned with all aspects of (business) local area networks, ranging from their performance to the specification and requirements of the supporting infrastructure. More recently he has become very involved in studies concerning the distribution of network delivered broadband services within the residential and small business environments. He chairs the ATM Forum Residential Broadband (RBB) Working Group, which is defining specifications for the delivery and distribution of ATM-based residential services.